

Partow Roshani

Computer Science Undergraduate

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Profile

Computer Science undergraduate at Shahid Beheshti University with interests in **Machine Learning, Data Science, Digital Image Processing, Algorithms, and Software Development**. Experienced in Python, C++, and Java through academic and applied projects spanning predictive maintenance, deep learning, graph algorithms, concurrent programming, and client-server software. Teaching Assistant with experience in programming instruction, assignment design, individual mentoring, and software project supervision.

Education

Shahid Beheshti University

Tehran, Iran Sep. 2023 – Present

B.Sc. in Computer Science

- **Cumulative GPA: 18.67/20**
- Relevant coursework: Machine Learning, Data Science, Digital Image Processing, Data Structures & Algorithms, Concurrent Programming, Advanced Programming, and Basic Programming.
- Selected grades: Advanced Programming **20/20**; Digital Image Processing **20/20**; Concurrent Programming **20/20**; Basic Programming **19.75/20**.

Alghadir High School

Kish, Iran Jul. 2023

Diploma in Mathematics and Physics

- Graduated with GPA **19.84/20**.

Teaching Experience

Shahid Beheshti University

Tehran, Iran Feb. 2026 – Aug. 2026

Teaching Assistant – Advanced Programming

- Served as a Teaching Assistant for the **Advanced Programming** course under Dr. Saeed Reza Kheradpisheh.
- Taught **Object-Oriented Programming (OOP)** concepts and supported students in applying object-oriented design principles in Java.
- Designed a programming assignment focused on **OOP concepts**, including class design, encapsulation, inheritance, polymorphism, and software structure.
- Mentored **three students individually**, providing regular support with programming concepts, debugging, problem solving, and code organization.
- Mentored **two student teams during their final projects**, advising them on software architecture, implementation decisions, debugging, and overall project development.

Shahid Beheshti University

Tehran, Iran Sep. 2025 – Feb. 2026

Teaching Assistant – Introduction to Programming

- Supported first-year Computer Science students in introductory programming and fundamental problem-solving techniques.
- Taught and assisted with core programming topics including **data types, control flow, loops, and functions**.
- Designed and reviewed programming exercises and helped students improve debugging, algorithmic thinking, and code organization.

Alghadir High School

Kish, Iran Nov. 2021 – Jan. 2022

Teaching Assistant – Physics and Chemistry

- Assisted students with coursework and problem solving in Physics and Chemistry.

Selected Projects

Jet Engine Hospital – Predictive Maintenance

Machine Learning

- Built an end-to-end predictive-maintenance system using **NASA C-MAPSS FD001, FD003, and FD004** turbofan-engine datasets.
- Developed pipelines for **Remaining Useful Life (RUL) regression**, failure-risk classification at 10, 20, and 30-cycle horizons, anomaly detection, and uncertainty estimation.
- Applied leakage-safe engine-level splitting, temporal feature engineering, operating-condition-aware processing, probability calibration, and conformal prediction intervals.

- Integrated model outputs into maintenance decision policies and developed an interactive **Streamlit dashboard** for engine-level inference and visualization.
- **Tech:** Python, Pandas, NumPy, Scikit-learn, Matplotlib, Plotly, Streamlit.

Deep Image Restoration and Classification

Digital Image Processing / Deep Learning

- Developed an end-to-end deep-learning pipeline for image denoising, deblurring, model comparison, and semantic image classification.
- Evaluated **CNN, ResNet, and U-Net** architectures for image deblurring using PSNR and automatically selected the best-performing model.
- Trained a **PreActResNet18** classifier on CIFAR-10, achieving **87.11% test accuracy** and **0.8897 validation macro F1-score**.
- Worked with both **TensorFlow and PyTorch** within the same image-processing pipeline.
- **Tech:** Python, PyTorch, TensorFlow, NumPy, Matplotlib.

Cost-Sensitive Customer Churn Prediction

Data Science

- Built an end-to-end customer churn prediction pipeline including data quality analysis, cleaning, feature engineering, preprocessing, class-imbalance handling, and model evaluation.
- Benchmarked traditional and ensemble classifiers including Logistic Regression, Random Forest, SVM, XGBoost, LightGBM, CatBoost, and Stacking.
- Designed a **cost-sensitive decision strategy** and optimized the classification threshold according to a custom business utility function.
- Used **SHAP** to analyze feature importance and explain model predictions.
- **Tech:** Python, Pandas, Scikit-learn, XGBoost, LightGBM, CatBoost, SHAP.

Personalized PageRank for Fraud Detection

Data Structures & Algorithms

- Implemented a Personalized PageRank framework for large-scale graph analysis using a performance-oriented **C++17 backend**.
- Implemented Power Iteration, deterministic seed selection, convergence monitoring, Precision@K evaluation, and scalability experiments.
- Designed batch experiments across datasets and algorithm parameters and generated structured results for comparative analysis.
- Built a Python/Tkinter graphical interface that communicates with the C++ backend through structured JSON output.
- **Tech:** C++, Python, Graph Algorithms, Tkinter, JSON, CSV.

Telegram Clone

Advanced Programming

- Developed a real-time messaging application using a client-server architecture with **Java, JavaFX, PostgreSQL, and socket programming**.
- Implemented multithreaded server-side connection handling and structured JSON communication between clients and the server.
- Supported private chats, saved messages, groups, channels, message persistence, and delivery/read states.
- Worked on GUI development, database integration, concurrency, application logic, and client-server communication.
- **Tech:** Java, JavaFX, PostgreSQL, Socket Programming, Multithreading, JSON.

Technical Skills

Programming Languages:	Python, C++, Java, SQL, MATLAB
Machine Learning / Data:	Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, XGBoost, LightGBM, CatBoost, SHAP
Visualization:	Matplotlib, Plotly
Software Development:	Object-Oriented Programming, Data Structures & Algorithms, Concurrent Programming, Multithreading, Socket Programming
Databases:	PostgreSQL
Tools:	Git, GitHub, Jupyter Notebook, VS Code, Streamlit, JavaFX, Scene Builder

Achievements

Iran National University Entrance Examination (Konkur): Regional Rank 198 /41,279	Iran
National Rank 2,742 /147,317	2023

Languages

Persian: Native

English: Intermediate – IELTS Academic **5.5**